

Secure File Storage and Access Control

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CHAPTER 1: INTRODUCTION

1. Presentation of Institute

1.1. About TGI

TUX Global is a group of companies in Japan that has an initiative to contribute to the development of human resources in technology, to pay courtesy to Cambodia, and to strengthen the relationship between the two countries to create values and opportunities for both parties. The great initiative has reached senior Japanese and Cambodian officials with high locality. Thereafter, with the support from the Ministry of Education, Youth and Sport (MoEYS), funders, Japanese supporters, board of directors, and advisors as well as a shared vision and value of having human capital in technology and soft-skill in this fast evolution era of both countries, TUX Global Institute (TGI) was founded and established in 2020 having **Mr. YUSUNOKI SHOJI** as the Chairman, while **Mr. Ham Sokly** holds the position of President, and His Excellency SUSUMU Tai is the Honorary President.

1.2. Vision

To provide high-quality education to Cambodian youth, to develop professional human resources in Cambodia and around the world, and to become one of the top educational institutions in Cambodia.

1.3. Mission

To bring innovation to Cambodia's IT schools to raise the standard of education for all Cambodian youth.

1.4. Values

Own it - Do the right thing - Share your humanity.



Figure 1: Logo of TUX Global Institute

2. Presentation of Project

The Secure File Storage with Access Control project is about creating a safe place to store important files while controlling who can access them. It ensures that only authorized people can view, edit, or manage the files, keeping sensitive data protected. It also keeps a record of who accessed the files, providing transparency and accountability. This project aims to make file storage both secure and easy to use, protecting information from being misused or stolen.

2.1. Problem Statement

The main problem addressed by the project is the risk of unauthorized access and misuse of sensitive data. Many existing file storage systems lack proper security measures, making them vulnerable to data breaches, insider threats, and external attacks. Without robust access controls, users may gain more privileges than necessary, leading to potential data leaks or accidental changes.

2.2. Objective

According to the problem, it requires to create a secure system for storing and managing files, ensuring that only authorized users can access them. By doing some research, it comes up with some ideas to archive the main propose of the project:

- To manage communication (Client to File storage)
- To manage user access (Domain Controller)
- To control file permission

2.3. Solution

To meet the objectives mentioned above, the system need to integrate with DC for managing domain users, this project ensures centralized, role-based access control, reducing the risks associated with excessive privileges and unauthorized access. Additionally, the research has come up with the solution as the following:

- Implementing the Firewall (Communication management)
- Installing the Window Server (Use as domain controller)
- Configuring the Server (Use as file storage)

2.4. Hardware and Operating System Used

In this project, I have used various hardware and operating system as following:

- Hardware
 - Laptop
- Operating system
 - Ubuntu Server 22.04
 - Window 10 Pro
 - Window Server 2016
 - pfSense

CHAPTER 2: CONCEPT AND ANALYSIS

1. Pros and Cons

1.1. Advantages

- **Enhanced Security:** Centralized control using a Domain Controller ensures robust authentication and role-based access, reducing unauthorized access risks.
- **Scalability:** Easily accommodates growing numbers of users and files, making it suitable for organizations of any size.
- **Accountability:** Logging and monitoring features provide clear records of user activities, aiding in audits and incident response.
- **Data Protection:** Encryption ensures that data remains secure, even if the physical storage device is compromised.
- **User Convenience:** Centralized management simplifies user authentication and permission updates.

1.2. Disadvantages

- **Complex Setup:** Configuring the Domain Controller, file server, and access permissions can be intricate, increasing the likelihood of errors during setup.
- **Risk of Misconfiguration:** Improperly set permissions or policies could lead to unauthorized access or prevent legitimate users from accessing files, compromising both security and usability.
- **Resource Intensive:** The solution requires significant hardware and technical expertise to operate effectively, which may strain smaller organizations.
- **Single Point of Failure:** Reliance on the Domain Controller means that its downtime could disrupt access for all users.

2. Choice of Technologies

In this session, we reviewed the existing technology, and concepts are fit and usable for the requirements of the project. All technologies used and involved in the whole project are detailed as below:

2.1. VMware Workstation

VMware Workstation is a powerful virtualization software designed for desktop systems, enabling users to run multiple operating systems as virtual machines (VMs) on a single physical computer. It is widely used by IT professionals, developers, and enthusiasts for testing, development, training, and creating isolated environments for secure workloads. VMware Workstation supports various operating systems, including Windows, Linux, and macOS (as a guest), offering robust features for managing and configuring VMs.

Key Features of VMware Workstation:

- Multi-OS Support:
 - Runs multiple operating systems (e.g., Windows, Linux, BSD) simultaneously on a single machine.
 - Enables cross-platform compatibility testing and development.

- Snapshots and Cloning:
 - Snapshots: Capture the state of a VM, allowing users to revert to a previous point easily.
 - Cloning: Duplicate VMs for faster deployment or testing environments.
- Networking Capabilities:
 - Supports various network configurations, such as NAT, Bridged, and Host-Only networks, to simulate complex network setups.
 - Enables VLAN configurations and integration with virtual firewalls like pfSense.
- Shared Folders and Drag-and-Drop:
 - Facilitates file sharing between the host and guest OS through shared folders.
 - Allows drag-and-drop functionality for easy file transfers.
- High-Performance Virtualization:
 - Leverages hardware acceleration technologies like Intel VT-x and AMD-V for efficient VM performance.
 - Supports up to 32 CPUs, 128GB of RAM, and 8TB virtual disks per VM, depending on the host system.
- Advanced Security:
 - Offers features like VM encryption, Secure Boot, and UEFI firmware support for secure virtualization.
- Integration with Cloud Platforms:
 - Integrates with VMware vSphere and other cloud solutions, enabling easy migration of workloads between local and cloud environments.
- Support for Containers and Kubernetes:
 - Offers Docker and Kubernetes compatibility for containerized development workflows.
- Cross-Platform Compatibility:
 - Available on Windows and Linux hosts, enabling users to choose their preferred environment.



Figure 2: VMware Workstation

2.2. pfSense

pfSense is an open-source firewall and router software based on FreeBSD, widely used for network management and security. It provides robust, enterprise-level features for managing network traffic, securing connections, and controlling access without the cost of proprietary solutions. pfSense is highly customizable, scalable, and user-friendly, making it suitable for businesses, educational institutions, and personal use. It can be deployed on dedicated hardware, virtual machines, or cloud platforms, and supports a wide range of functionalities beyond basic firewalling.

Key Features of pfSense:

- Firewall and NAT:
 - Stateful firewall with advanced rule sets to control traffic.
 - Network Address Translation (NAT) for mapping internal addresses to public IPs.
- VLAN Support:
 - Segregation of networks using Virtual LANs (VLANs) to isolate devices and enhance security.
- VPN Services:
 - Built-in support for OpenVPN, IPsec, and WireGuard to establish secure remote access and site-to-site connections.
- Intrusion Detection and Prevention:
 - Integration with Snort or Suricata for real-time monitoring and blocking of malicious activities.
- Traffic Shaping and Bandwidth Management:
 - Quality of Service (QoS) features to prioritize critical applications and manage bandwidth usage.
- High Availability:
 - Failover and redundancy configurations ensure continuous network operation.
- Logging and Reporting:
 - Detailed logs and graphical reports for monitoring traffic, firewall events, and system performance.
- Load Balancing:
 - Distributes traffic across multiple WAN or LAN connections for improved reliability and performance.
- Package System:
 - Expand functionality with additional packages like pfBlockerNG (for ad-blocking and geofencing), Squid (proxy server), and more.
- Web-Based Interface:
 - Easy-to-use GUI for configuration and monitoring, accessible via a web browser.
- Dynamic DNS:
 - Support for dynamic DNS services to update domain records automatically.
- Cost Efficiency:
 - Completely free to use with community support, offering an affordable alternative to commercial firewall solutions.



Figure 3: pfSense

2.3. Window Server

Windows Active Directory (AD) is a directory service developed by Microsoft that is used to manage users, computers, and other resources in a network environment. Here's an overview of some key features and benefits of Active Directory:

- **Centralized management:** Active Directory provides a centralized location for managing user accounts, computer accounts, and other resources in a network environment.
- **Security:** Active Directory allows administrators to enforce security policies and access controls for network resources, ensuring that only authorized users have access to sensitive data and systems.
- **Scalability:** Active Directory is designed to scale to meet the needs of small and large organizations, with support for thousands of users and computers.
- **Group Policy:** Active Directory includes Group Policy functionality, which allows administrators to define and enforce policies and settings for users and computers in a network environment.
- **Integration with other Microsoft products:** Active Directory is tightly integrated with other Microsoft products, such as Exchange Server, SharePoint, and Office 365, providing a seamless user experience across multiple applications.

In Conclusion, Active Directory is a powerful tool for managing network resources in a Windows-based environment, providing centralized management, enhanced security, scalability, and integration with other Microsoft products.



Figure 4: Active Directory

2.4. Ubuntu Live Server

Ubuntu Live Server is an excellent choice for setting up a file-sharing server due to its stability, flexibility, and support for various protocols. It enables secure and efficient

file sharing across different platforms, such as Windows, macOS, and Linux. With tools like Samba for SMB/CIFS (common in Windows networks) and NFS for Linux/Unix environments, Ubuntu Live Server can handle diverse file-sharing needs. Its lightweight nature and command-line interface make it resource-efficient, allowing it to serve files reliably even on modest hardware. Additionally, built-in security features and regular updates ensure data protection and integrity.

Key Features of Ubuntu Live Server:

- Minimal and Lightweight:
 - Focuses on server-specific packages and tools, avoiding unnecessary bloat.
 - Optimized for better performance and resource efficiency.
- Command-Line Interface (CLI):
 - Offers a CLI for managing the server, ideal for advanced users and remote administration.
 - Lightweight and flexible, suitable for automation and scripting.
- Customizable Installation:
 - Provides a guided setup with options to install essential services like SSH, LAMP stack, Samba, or NFS.
 - Allows users to tailor the server for specific purposes.
- Built-in Network Configuration:
 - Simplifies setup with tools for configuring network interfaces and static or dynamic IP addresses.
- Cloud Integration:
 - Prepares servers for cloud environments with tools like cloud-init for initializing cloud instances.
- Security and Updates:
 - Regular security patches and updates to keep the server secure.
 - Features like unattended upgrades ensure that the server remains up-to-date.
- Long-Term Support (LTS):
 - Offers LTS versions with five years of support, ensuring stability and reliability for production environments.
- Built-in Snap Support:
 - Includes Snap for managing and running containerized applications.
 - Simplifies software installation and updates.
- Versatile Use Cases:
 - Can be configured as a file server, web server, DNS server, or database server.
 - Suitable for virtualization and containerized deployments (e.g., using Docker).
- Live Mode:
 - Allows you to boot and run Ubuntu Live Server without installation for troubleshooting or testing.



Figure 5: Ubuntu

2.5. Window Pro Client

Windows Pro is a professional-grade operating system from Microsoft, designed for business and professional environments. It offers advanced features for networking, security, and productivity, making it ideal for client systems in server-based networks. Windows Pro seamlessly integrates with enterprise tools like Active Directory (via Domain Controllers), file sharing, and remote access, allowing users to work efficiently in managed IT environments.

Key Features of Windows Pro as a Client:

- Domain Join and Active Directory Integration:
 - Allows the client to join a domain for centralized user authentication and management.
 - Facilitates role-based access control to shared files and resources on the network.
- File Sharing and Network Access:
 - Supports file sharing protocols like SMB for seamless interaction with file servers.
 - Enables mapping of network drives for easy access to shared folders.
- Remote Desktop Access:
 - Includes Remote Desktop Protocol (RDP) for accessing the system remotely.
 - Ideal for remote troubleshooting and work-from-home scenarios.
- BitLocker Drive Encryption:
 - Protects local data with encryption, ensuring sensitive files remain secure in case of device theft or loss.
- Group Policy Management:
 - Supports centralized control of system settings via Group Policy, enabling administrators to enforce security and configuration policies.
- Enhanced Security Features:
 - Includes Windows Defender and advanced firewall capabilities for robust protection against malware and unauthorized access.
- Windows Update for Business:
 - Offers flexible update management, ensuring clients stay updated while minimizing disruptions.

- Professional Productivity Tools:
 - Preloaded with tools like Microsoft Edge and compatibility with Office applications to enhance productivity.



Figure 6: Window 10 Pro

3. Diagram and Network Architecture

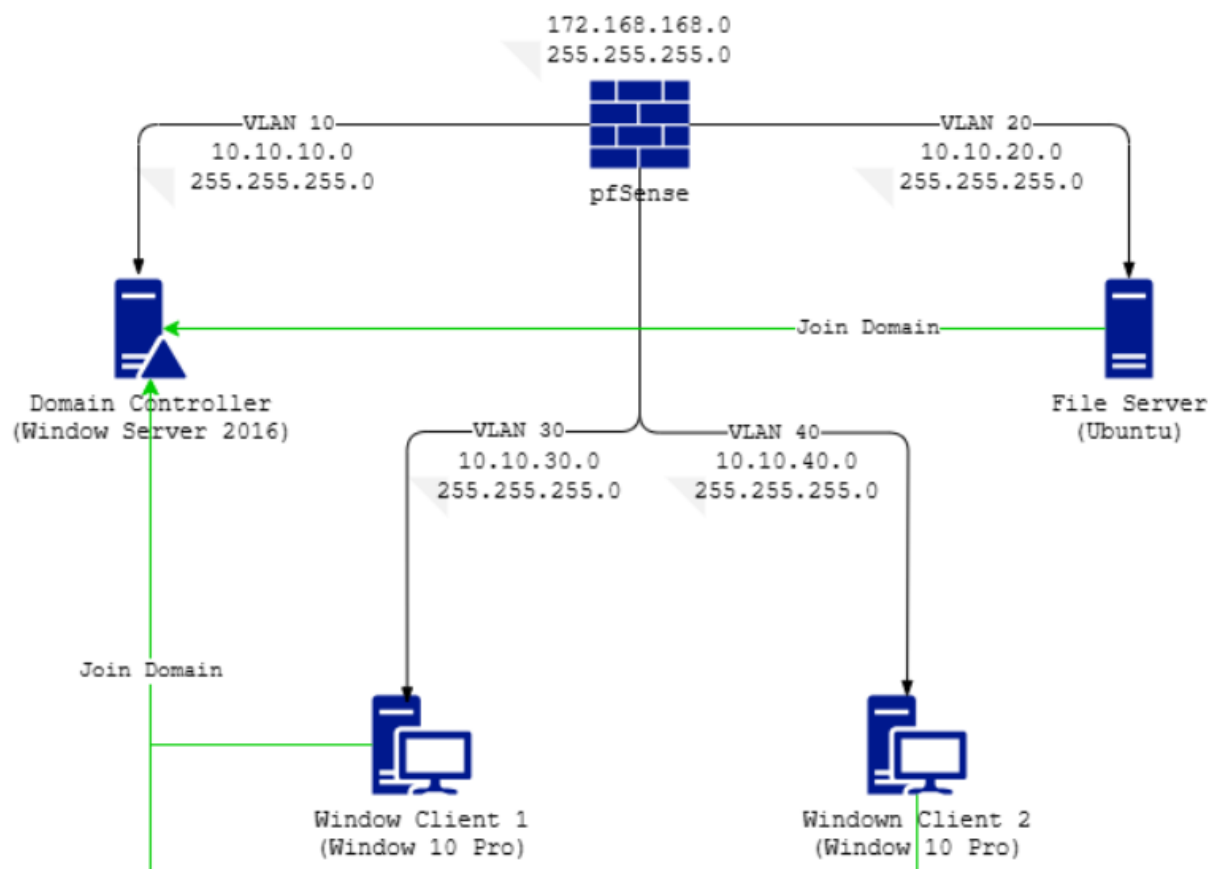


Figure 7: Diagram and Network architecture

At the core is a pfSense firewall, which manages traffic and communication between multiple VLANs. The network is divided into subnets for logical isolation: VLAN 10 hosts a Domain Controller (Windows Server 2016) responsible for centralized authentication and domain management, VLAN 20 contains an Ubuntu-based file server for secure file sharing, and VLANs 30 and 40 connect two Windows 10 Pro clients. Both clients and file server are joined domain as domain member, allowing them to access shared resources on the file server with credentials managed by the Domain Controller. VLAN segmentation enhances security by isolating devices, while pfSense enables controlled inter-VLAN communication. This setup ensures streamlined authentication, centralized user management, and secure resource sharing.

CHAPTER 3: IMPLEMENTATION

In the session, we are going to detail and describe the main configuration of our project. As we are working on our own laptop, we implemented it with minimum resources due to the limited resources of our laptop. Otherwise, the project is completely done and aligned with our scope.

1. pfSense

1.1. System Requirement

Our environment	Best practice
RAM: 2GB	RAM: 4GB
HDD/SSD: 20GB	HDD/SSD: 20GB
CPU: 2 cores and 2 processors	CPU: 2 cores and 2 processors

1.2. Setting up

We have set up 5 loopback adapters for VLAN segmentation.

```

WAN (wan)      -> le0      -> v4/DHCP4: 192.168.1.210/24
LAN (lan)      -> le1      -> v4: 172.168.168.168/24
AD (opt1)      -> le2      -> v4: 10.10.10.1/24
CLIENT1 (opt2) -> le3      -> v4: 10.10.30.1/24
CLIENT2 (opt3) -> le4      -> v4: 10.10.40.1/24
FILESERVER (opt4) -> le5    -> v4: 10.10.20.1/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults  13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                15) Restore recent configuration
7) Ping host                  16) Restart PHP-FPM
8) Shell

Enter an option: █

```

Figure 8: pfSense network adapters

We configured some rules for communication between each machine.

Floating WAN LAN **AD** CLIENT1 CLIENT2 FILESERVER

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 0/0 B	IPv4 TCP/UDP	AD net	*	FILESERVER net	FileServer_DC_TCP_ UDP	*	none		Allow DC to respond to File Server requests	
☐	✓ 0/0 B	IPv4 TCP/UDP	AD net	*	CLIENT1 net	DC_Client_TCP_UDP	*	none		Allow DC to respond to Windows Client requests	
☐	✓ 0/0 B	IPv4 TCP/UDP	AD net	*	CLIENT2 net	DC_Client_TCP_UDP	*	none		Allow DC to respond to Windows Client requests	
☐	✓ 0/0 B	IPv4 ICMP echo req	AD net	*	FILESERVER net	*	*	none		Allow ICMP from DC to File Server	
☐	✓ 0/0 B	IPv4 TCP/UDP	AD net	*	*	InternetPorts	*	none			

Add
 Add
 Delete
 Save
 Separator

Figure 9: AD rules

Floating WAN LAN **AD** CLIENT1 CLIENT2 FILESERVER

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 0/0 B	IPv4 TCP/UDP	FILESERVER net	*	AD net	FileServer_DC_TCP_ UDP	*	none		Allow File Server to authenticate and communicate with DC	
☐	✓ 0/0 B	IPv4 ICMP echo req	FILESERVER net	*	AD net	*	*	none		Allow ICMP from File Server to DC	
☐	✓ 0/0 B	IPv4 TCP/UDP	FILESERVER net	*	*	InternetPorts	*	none			

Add
 Add
 Delete
 Save
 Separator

Figure 10: Fileserver rules

Floating WAN LAN **AD** **CLIENT1** CLIENT2 FILESERVER

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 0/0 B	IPv4 TCP	CLIENT1 net	*	FILESERVER net	Client_FileServer_ TCP	*	none		Allow Windows Clients to access File Server	
☐	✓ 0/0 B	IPv4 TCP/UDP	CLIENT1 net	*	AD net	DC_Client_TCP_ UDP	*	none		Allow Windows Clients to authenticate and query DC	
☐	✓ 0/0 B	IPv4 ICMP echo req	CLIENT1 net	*	AD net	*	*	none			
☐	✓ 0/0 B	IPv4 ICMP echo req	CLIENT1 net	*	FILESERVER net	*	*	none		Allow ICMP from Windows Clients to File Server	

Add
 Add
 Delete
 Save
 Separator

Figure 11: Client rules

We also enabled the DHCP server to each interface for providing an IP to the machine.

Services / DHCP Server / AD

LAN **AD** CLIENT1 CLIENT2 FILESERVER

General Options

Enable ☒ Enable DHCP server on AD interface

BOOTP ☐ Ignore BOOTP queries

Deny unknown clients Allow all clients

When set to **Allow all clients**, any DHCP client will get an IP address within this scope/range on this interface. If set to **Allow known clients from any interface**, any DHCP client with a MAC address listed on **any** scope(s)/interface(s) will get an IP address. If set to **Allow known clients from only this interface**, only MAC addresses listed below (i.e. for this interface) will get an IP address within this scope/range.

Ignore denied clients ☐ Denied clients will be ignored rather than rejected.
This option is not compatible with failover and cannot be enabled when a Failover Peer IP address is configured.

Ignore client identifiers ☐ If a client includes a unique identifier in its DHCP request, that UID will not be recorded in its lease.
This option may be useful when a client can dual boot using different client identifiers but the same hardware (MAC) address. Note that the resulting server behavior violates the official DHCP specification.

Subnet 10.10.10.0

Subnet mask 255.255.255.0

Available range 10.10.10.1 - 10.10.10.254

Range 10.10.10.20 10.10.10.100
From To

Additional Rules

Figure 12: Sample DHCP configuration

Some necessary ports you need to create for allow each connectivity.

Firewall / Aliases / Ports

IP **Ports** URLs All

Firewall Aliases Ports

Name	Values	Description	Actions
Client_FileServer_TCP	445, 135		
DC_Client_TCP_UDP	53, 88, 389, 445, 135, 3268, 137, 138, 139		
FileServer_DC_TCP_UDP	53, 88, 389, 445, 3268, 135, 464		
InternetPorts	80, 443, 53	Allow Access Internet	

Add Import

Figure 13: Aliases configuration

2. Windows Server 2016

2.1. System Requirement

Our environment

RAM: 4GB
HDD/SSD: 60GB
CPU: 2 cores and 2 processors

Best practice

RAM: 8GB
HDD/SSD: 100GB
CPU: 2 cores and 2 processors

2.2. Setting up

We have set up a domain, called "tgiproject.com.kh" and assigned the static IP "10.10.10.21".

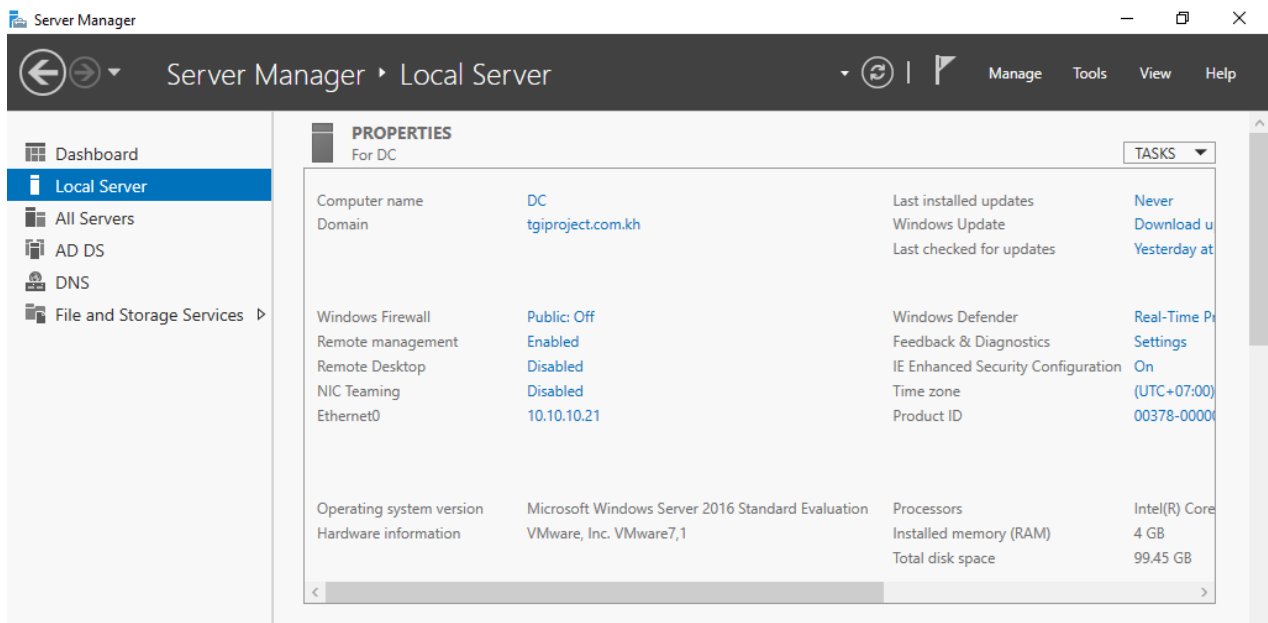


Figure 14: Active directory set up

We have joined these 3 machines to domain controller.

- Fileserver - Ubuntu server
- ITD - Window client (IT Department)
- IAD - Window client (IA Department)

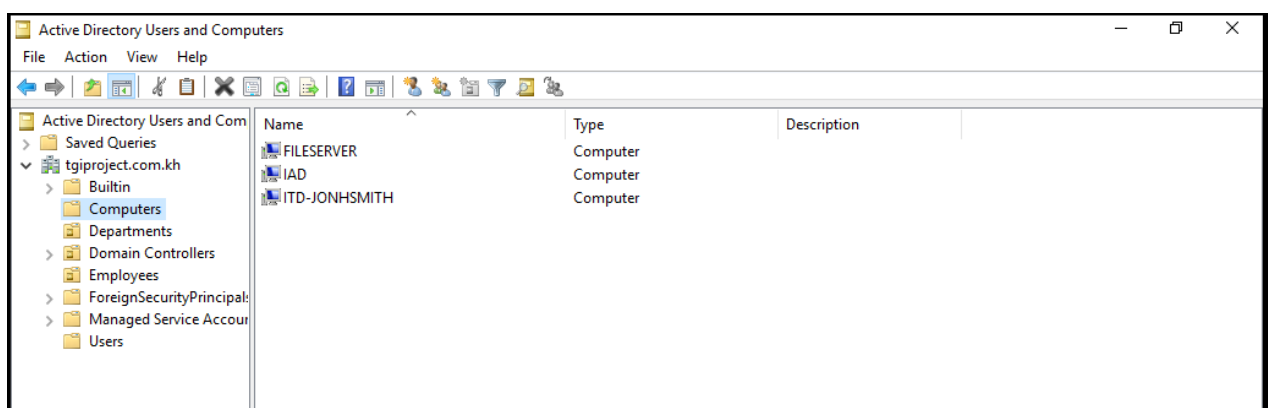


Figure 15: Joined domain computer

Create user for each department

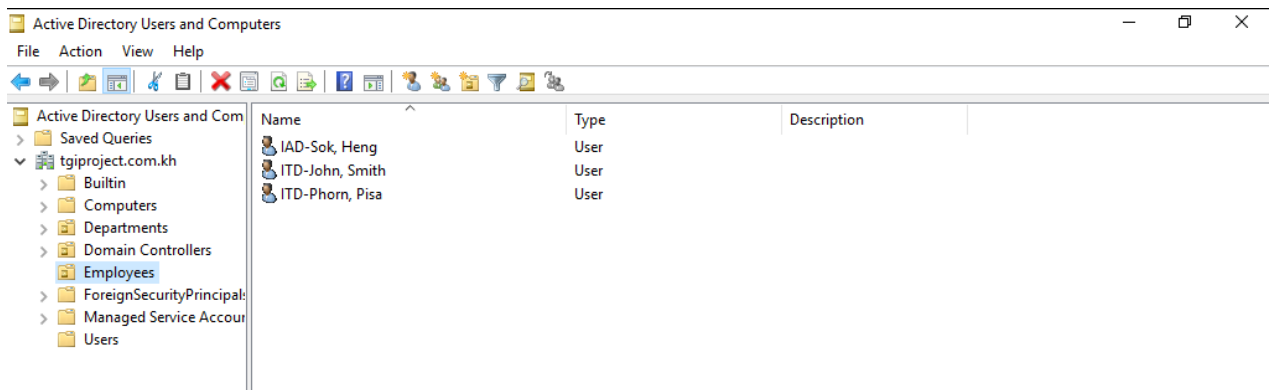


Figure 16: Domain users

Every user belongs to their department based on a domain group.

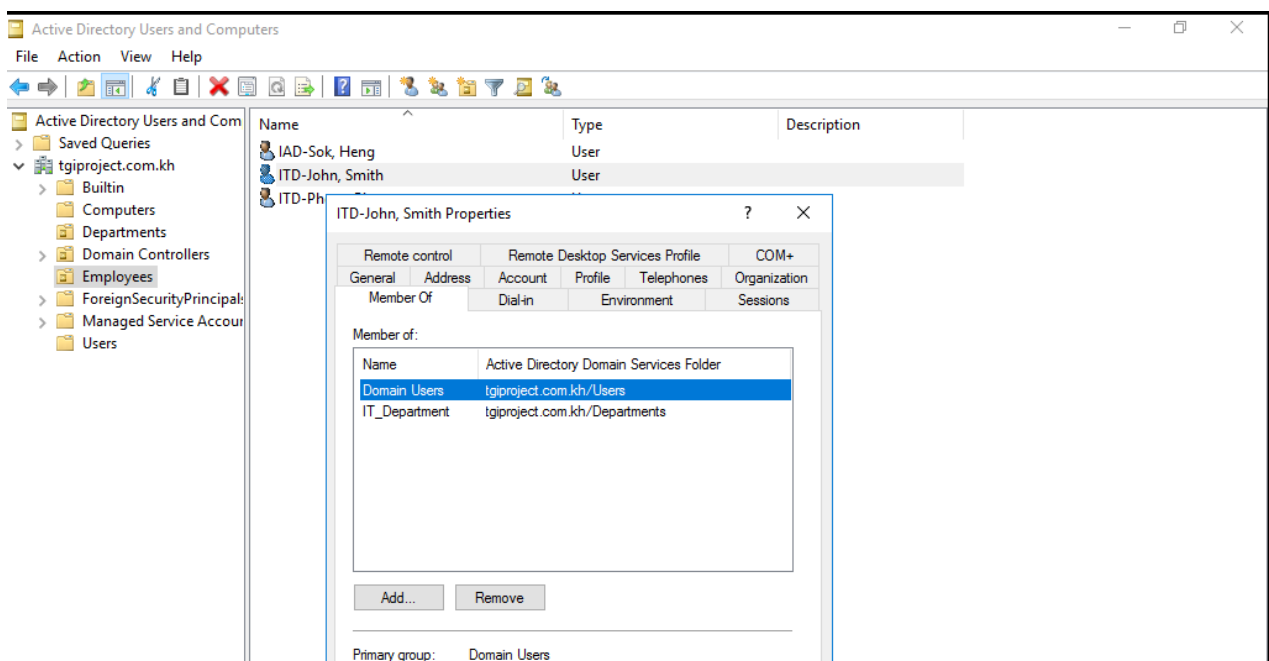


Figure 17: Domain user setting

3. Fileserver

3.1. System Requirement

Our environment

RAM: 2GB
HDD/SSD: 60GB
CPU: 2 cores and 2 processors

Best practice

RAM: 4GB
HDD/SSD: 100GB
CPU: 2 cores and 2 processors

3.2. Setting up

Installing the necessary packages.

```
# sudo apt update
```

```
# sudo apt install samba samba-common-bin realmd sssd sssd-tools adcli krb5-user ntp -y
```

Update /etc/hosts and /etc/resolv.conf to point to the DC for DNS resolution.

Edit /etc/hosts

```
# sudo nano /etc/hosts
```

Add:

```
10.10.10.21 dc.tgiproject.com.kh
```

```
10.10.20.26 fileserver.tgiproject.com.kh
```



```
GNU nano 7.2 /etc/hosts
127.0.0.1 localhost
10.10.20.26 fileserver.tgiproject.com.kh fileserver
#Domain Controller
10.10.10.21 dc.tgiproject.com.kh dc
```

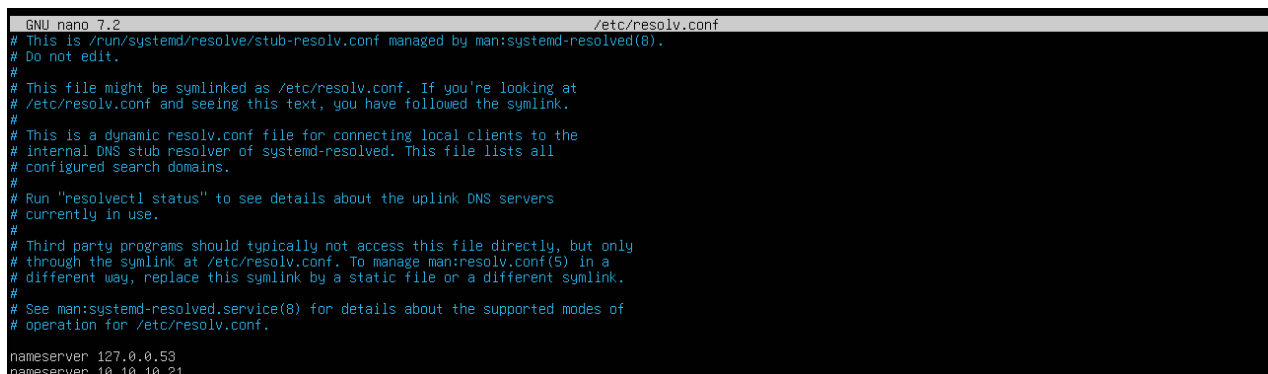
Figure 18: Host configuration

Edit /etc/resolv.conf:

```
# sudo nano /etc/resolv.conf
```

Add:

```
nameserver 10.10.10.21
```



```
GNU nano 7.2 /etc/resolv.conf
# This is /run/systemd/resolve/stub-resolv.conf managed by man:systemd-resolved(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients to the
# internal DNS stub resolver of systemd-resolved. This file lists all
# configured search domains.
#
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs should typically not access this file directly, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.
nameserver 127.0.0.53
nameserver 10.10.10.21
```

Figure 19: Resolv configuration

Discover and Join the Domain

```
# sudo realm discover tgiproject.com.kh
```

```
# sudo realm join --user=Administrator tgiproject.com.kh
```

Verify Domain Integration

```
# realm list
```

```

root@fileserver:~# realm list
tgiproject.com.kh
  type: kerberos
  realm-name: TGIPROJECT.COM.KH
  domain-name: tgiproject.com.kh
  configured: kerberos-member
  server-software: active-directory
  client-software: sssd
  required-package: sssd-tools
  required-package: sssd
  required-package: libnss-sss
  required-package: libpam-sss
  required-package: adcli
  required-package: samba-common-bin
  login-formats: %U@tgiproject.com.kh
  login-policy: allow-realm-logins
root@fileserver:~# _

```

Figure 20: Fileserver joined domain

Configure Samba for File Sharing

```
# sudo nano /etc/samba/smb.conf
```

```

GNU nano 7.2 /etc/samba/smb.conf
[global]
security = user
dedicated keytab file = /etc/krb5.keytab
log file = /var/log/samba/%m.log
log level = 1

[01_ITD]
path = /srv/shared/01_ITD
valid users = it
read only = no
browsable = yes

[02_IA0]
path = /srv/shared/02_IA0
valid users = ia
read only = no
browsable = yes

```

Figure 21: Samba configuration

We need to create and set password for smb user to grant permission for accessing to the following folder.

```

# sudo useradd it
# sudo useradd ia
# sudo smbpasswd -a it
# sudo smbpasswd -a ia
# sudo smbpasswd -e it
# sudo smbpasswd -e ia
# sudo systemctl restart smbd
# sudo systemctl enable smbd

```

And we need to allow firewall rules for samba as well

```

# sudo ufw allow samba
# sudo ufw status

```

```

root@fileserver:~# sudo ufw status
Status: active

To Action From
--
88/tcp ALLOW Anywhere
88/udp ALLOW Anywhere
389/tcp ALLOW Anywhere
389/udp ALLOW Anywhere
464/tcp ALLOW Anywhere
464/udp ALLOW Anywhere
Samba ALLOW Anywhere
88/tcp (v6) ALLOW Anywhere (v6)
88/udp (v6) ALLOW Anywhere (v6)
389/tcp (v6) ALLOW Anywhere (v6)
389/udp (v6) ALLOW Anywhere (v6)
464/tcp (v6) ALLOW Anywhere (v6)
464/udp (v6) ALLOW Anywhere (v6)
Samba (v6) ALLOW Anywhere (v6)

root@fileserver:~# _

```

Figure 22: Fileserver firewall status

Create the shared folder and set permissions for each folder:

```

# sudo mkdir -p /srv/shared/01_ITD
# sudo chown root:it_department /srv/shared/01_ITD
# sudo chmod 770 /srv/shared/01_ITD

```

Restart the Samba services:

```

# sudo systemctl restart smbd nmbd winbind

```

4. Window Client

We have joined domain for all window client:

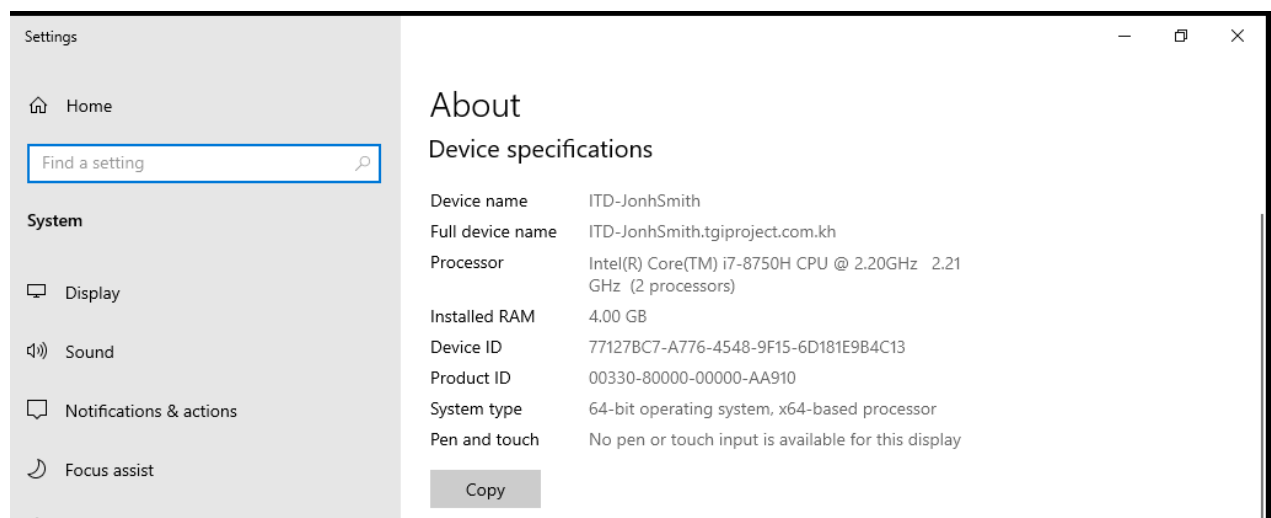


Figure 23: Computer has joined the domain

We can access to fileserver by entering:

\\fileserv or \\ip_of_fileserv

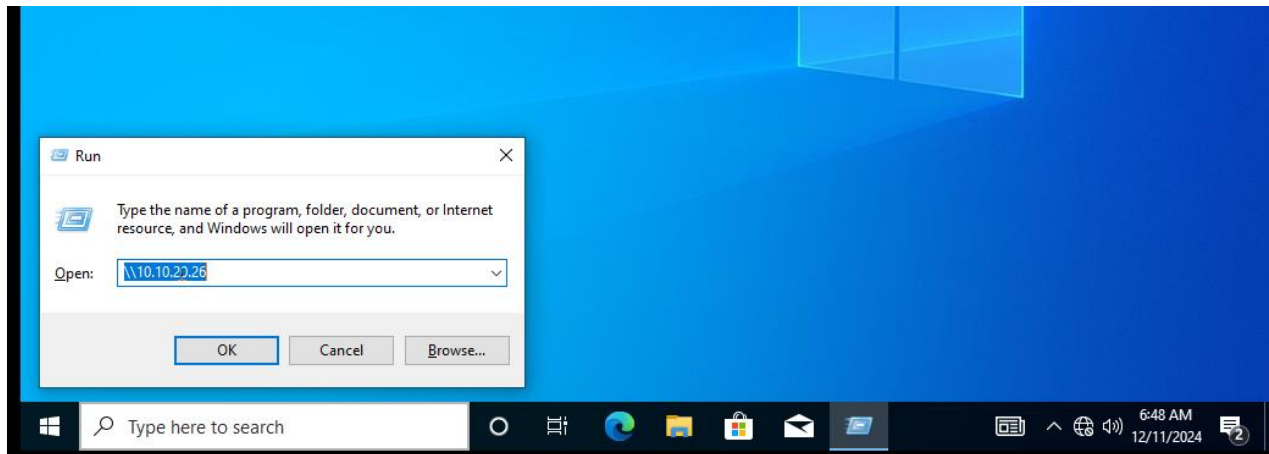


Figure 24: Access to fileserv

Enter the credentials of smb user you have created for each folder:

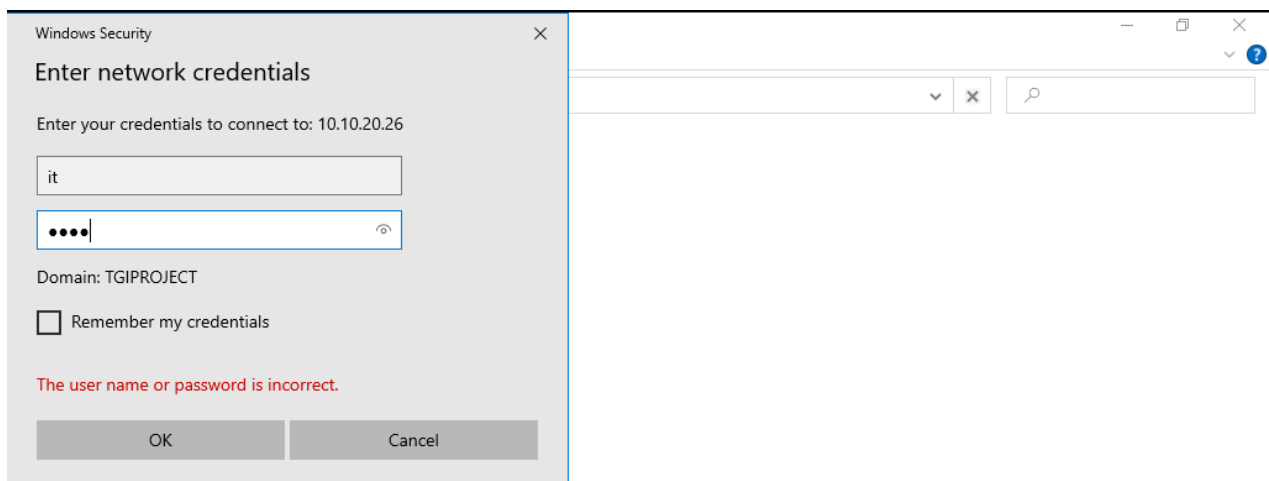


Figure 25: SMB Credentials requirement

Now you can access only the folder of your department:

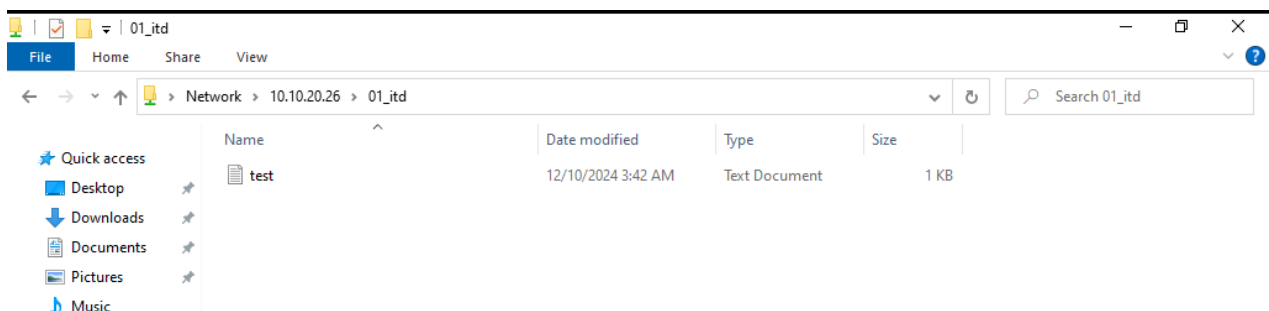


Figure 26: Folder accessible

Other folders will not be accessible:

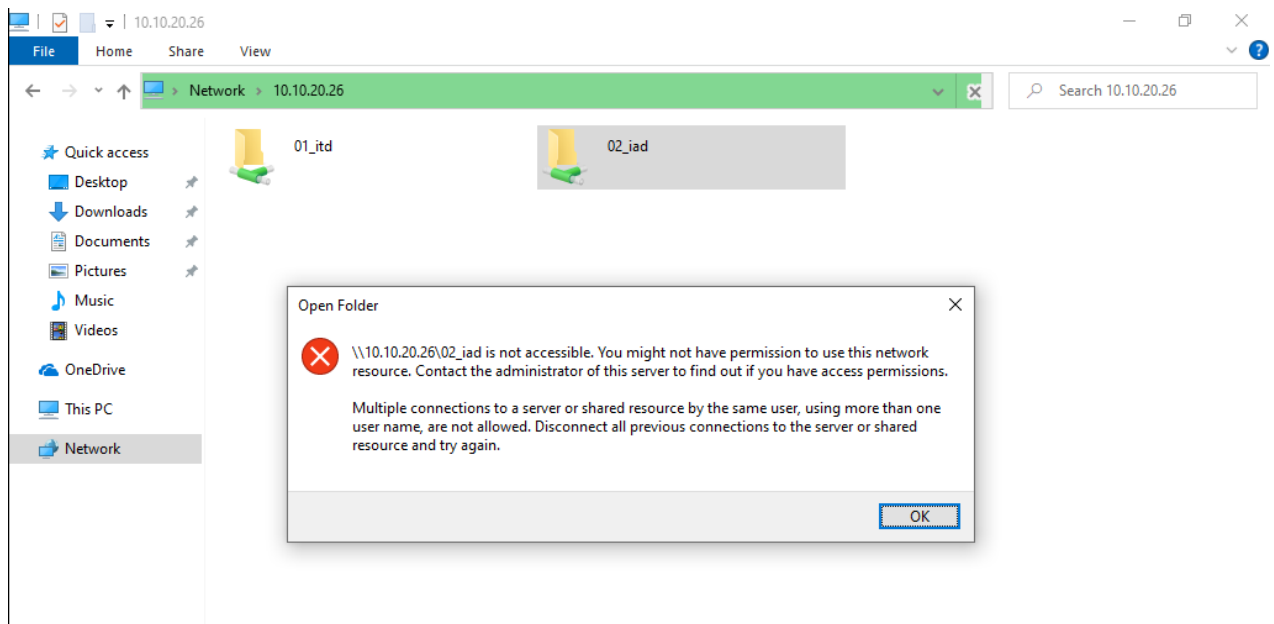


Figure 27: Inaccessible to another folder

Outside network also cannot access to each folder:

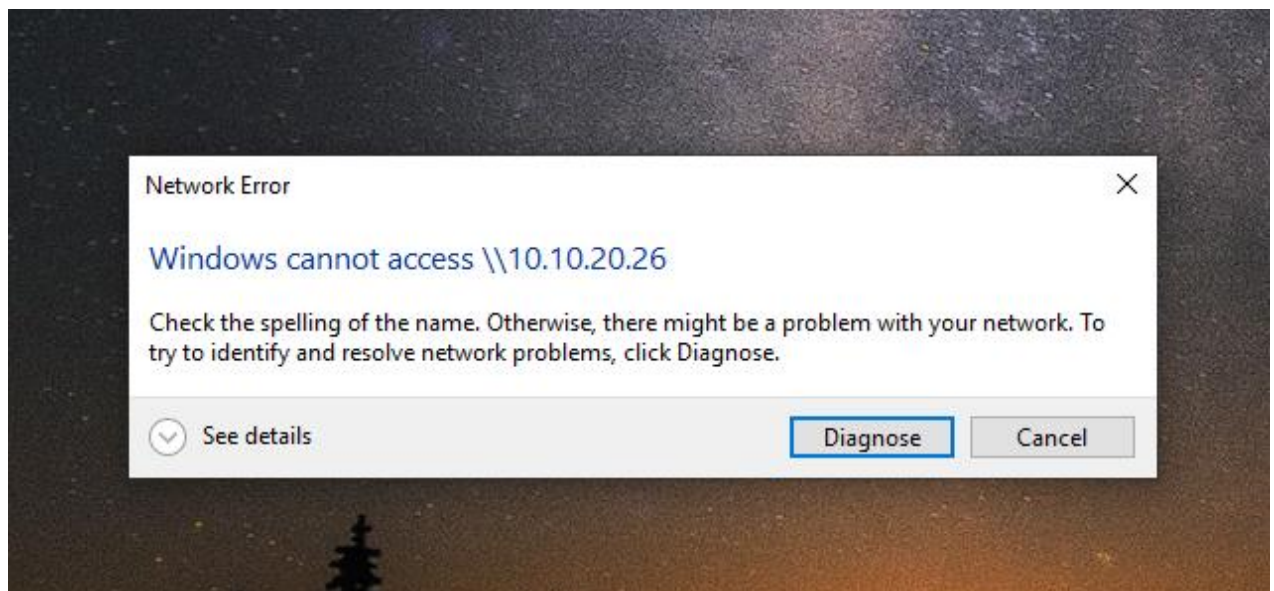


Figure 28: External network cannot access

CHAPTER 4: CONCLUSION

This is the final chapter of my report, where I summarize the work, I did for this project and share what I learned. I will also talk about the strengths, weaknesses, and challenges I faced while working on it.

1. General Conclusion

This project taught me a lot about setting up and managing a secure file storage system. I used Ubuntu as the file server, pfSense as a firewall, Windows Server 2016 as a domain controller for the "tgiproject.com.kh" domain, and two Windows client systems. Through this, I learned how to manage file sharing, secure a network, and control user access. This project helped me improve my problem-solving, time management, and technical skills. Overall, it was a valuable experience that will be helpful for my future work.

2. Strong Point

The biggest strength of this project was its practical application. I learned how to use Ubuntu for secure file storage, pfSense for protecting the network, and Windows Server 2016 for user management. Setting up and connecting everything taught me how to secure files and manage a system properly. This project gave me real-world skills and confidence in managing and securing IT systems.

3. Weak Point

One of the challenges was handling everything by myself. It was hard to manage all the tasks, especially since some technologies, like pfSense and Windows Server 2016, were new to me. This made the work take more time than expected. I also realized that working with a team could have made the process easier and faster by sharing ideas and tasks. In the future, I want to improve my teamwork and time management skills.